Filtering before testing

Using ld_w to choose test units rather than to rank them

module_sim_LDscnR

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Contents

1 What this document is

This describes a **different analysis from the C-score work**, resting on the same premise. It is written separately because the machinery, the unit of analysis, the truth definition and the comparison are all different, and reading the two as variants of one thing has already caused confusion.

The shared premise. ld_w is computed from genotypes alone. It never sees the phenotype. If it carries information about *where* causal variation sits, that information is available before any association test is run.

What changed. The C-score used that information to **rank** markers. This uses it to **choose which units to test**. Ranking failed. Selection works — with one substantial qualification developed in Section 6.

2 Why the ranking approach was abandoned

Recorded so this document is self-contained, and so the same ground is not re-covered. All comparisons are paired within run, with a sign test; never a difference of means.

Test	Result
$\tau_C = 0.05$ vs $\alpha = 0.05$	C better in 79/151 (52%), $p = 0.63$
PR-AUC at the operative distance cap	C better in 120/313 (38%), $p = 4.4 \times 10^{-5}$
Stage-2 clusters as fixed units	α ahead by 0.07 PR, flat at every size floor
Restricting $\rho \geq 0.5$	C worse by 0.002–0.009

Two external results agree. The 3sp panel session finds C *ties* an oracle best-single-grid-cell rather than beating it, so the consistency layer buys robustness to parameter choice, not power. The simulation-parsing session measured the mechanism directly: reweighting a p-value by ld_w moves a median 32% of the top-k and nets **about one marker in 1,400** at 50–100 kb, and exactly zero at 10 kb, which is tagging scale. A reweighting that small cannot produce a detectable ranking gain, so the null result and the mechanism agree.

3 The selection framing

3.1 Independent filtering

Selecting k units and applying BH within that set is **independent filtering** in the sense of Bourgon, Gentleman and Huber (2010). The multiplicity correction is applied over k tests rather than all of them, so the threshold is less severe. This is arithmetic — it would work for *any* filter — which is exactly why the informative comparisons are not against the unfiltered scan but against **other filters of the same size**.

Filtering is valid provided the filter statistic is independent of the test statistic under the null. ld_w is phenotype-blind, which is necessary but not sufficient: ld_w correlates with allele frequency, and frequency drives power. Section~4.4 tests this empirically rather than assuming it.

3.2 The machinery

Component	**Choice**	**Why**
Unit	Stage-2 LD cluster, represented by its pruned marker	Membership is recomputed per bundle and required to reproduce the stored GRM marker set exactly
Filter	Top k units by median ld_w of members, computed before any phenotype	Median rather than max: robust to a single extreme member
Test	BH at 0.05 within the selected set only	Identical to the conventional analysis, applied to fewer tests
Truth	bp distance to nearest driving QTN, causal variants excluded	Shares no machinery with ld_w; an r^2-tagging truth would flatter it by construction
Controls	MAF, cluster size, random — all at identical k	Isolates enrichment from mere size reduction, and ld_w from frequency
Null	Surrogate environments (genetic basis), 100 draws	Preserves spatial and genetic structure; a naive shuffle destroys what ld_w keys on

The genome-wide scan — BH at 0.05 over all units — **is** the conventional α analysis. It is the baseline in every comparison below, so no separate benchmark is needed.

4 Results

4.1 Power

Table 1: Percent of 80 panels in which each filter yields MORE true discoveries than the genome-wide alpha analysis. 50 = no better.

method	k	10 kb	50 kb	100 kb
ld_w	500	56	58	61
ld_w	1000	65	63	66
ld_w	2000	73	73	75
ld_w	5000	79	79	80
ld_w	10000	81	81	81
ld_w	20000	97	100	100
size	500	77	75	72
size	1000	81	78	76
size	2000	85	82	79
size	5000	91	82	80
size	10000	92	82	78
size	20000	91	81	81
MAF	500	8	8	9
MAF	1000	13	13	15
MAF	2000	16	18	21
MAF	5000	21	26	26
MAF	10000	29	29	29
MAF	20000	62	65	66
random	500	2	2	2
random	1000	4	2	2
random	2000	2	2	2
random	5000	2	2	2
random	10000	2	0	0
random	20000	4	2	2

Two things follow. ld_w and cluster size both rise well above 50% and keep rising with k; **MAF and random selection sit far below 50%**, meaning they are worse than not filtering. Filtering normally *costs* real signal, and only a filter enriched for causal neighbourhoods repays that cost.

The gain also scales with the multiplicity burden, as the mechanism predicts: on 30k-marker tracks ld_w reached 86% at $k = 5{,}000$, against 79%% on these ~86k-unit panels.

4.2 The mechanism

Filtering changes no p-value. It changes the BH threshold, because BH over 99,786 units is far more severe than BH over 5,000.

On that panel the count goes from 27 discoveries (16 true) to 54 (22 true). **Of the 28 gained units, 6 (21%) lie within 50 kb of a driving QTN**, so four in five gained discoveries are not near one. Precision falls as recall rises; that is the trade, and it is why Section~4.4 is not optional.

4.3 What the target actually looks like

Only **459 of 99,786 clusters** reach $r^2 \geq 0.2$ with any driving QTN — under 0.5%. The strongest signals are the QTN-carrying clusters themselves, while the surrounding neighbourhood at $r^2 = 0.3$–0.6 mostly sits below $-\log_{10}(p) = 2$: recovering the *neighbourhood* is much harder than recovering the variant. Several clusters that **carry** a causal variant sit at $p \approx 1$, so carrying one is no guarantee of detection.

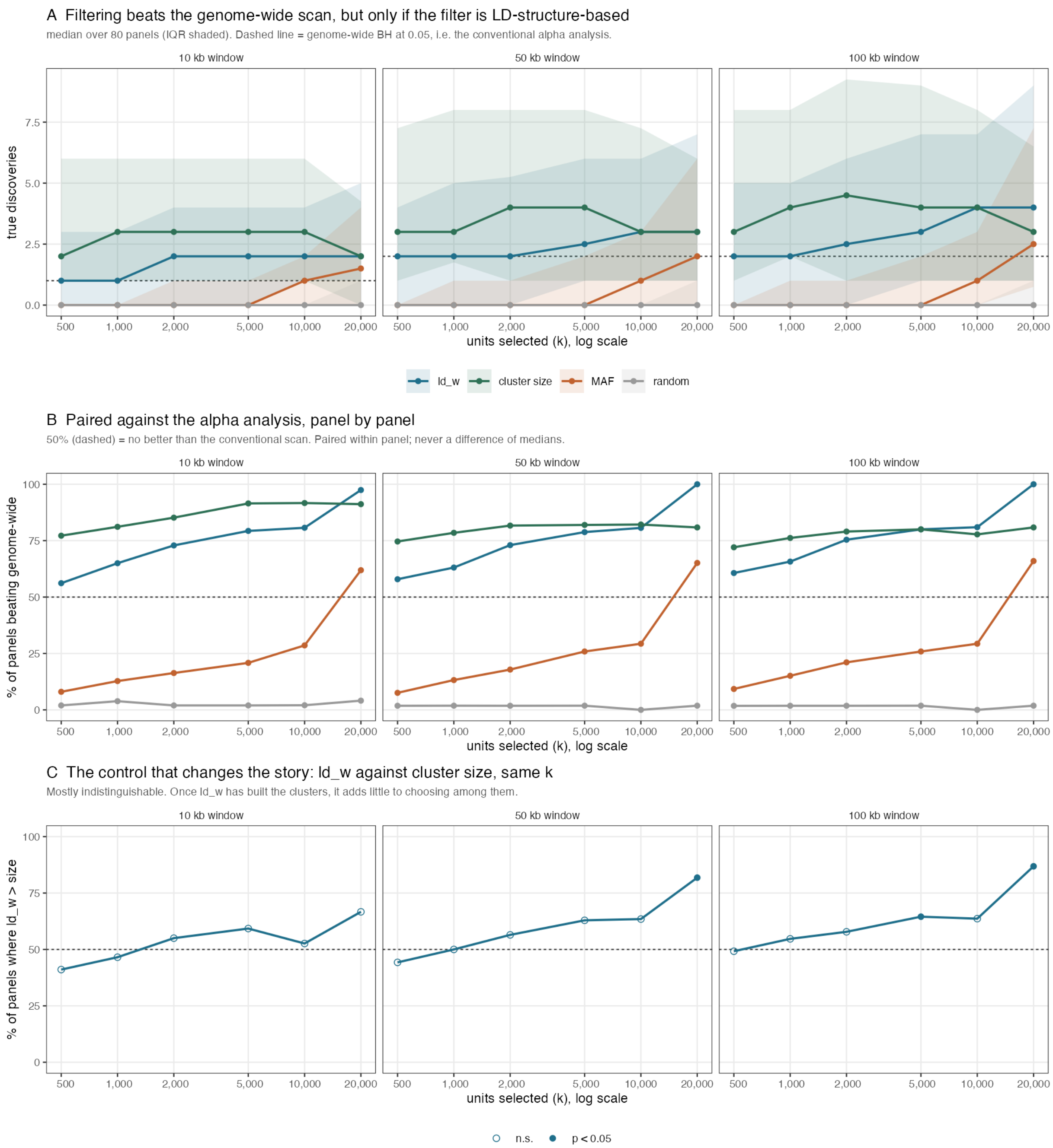

Figure 1: Filter-then-test against the conventional alpha analysis. Panel A: true discoveries. Panel B: paired win rate, the primary result. Panel C: the size control.

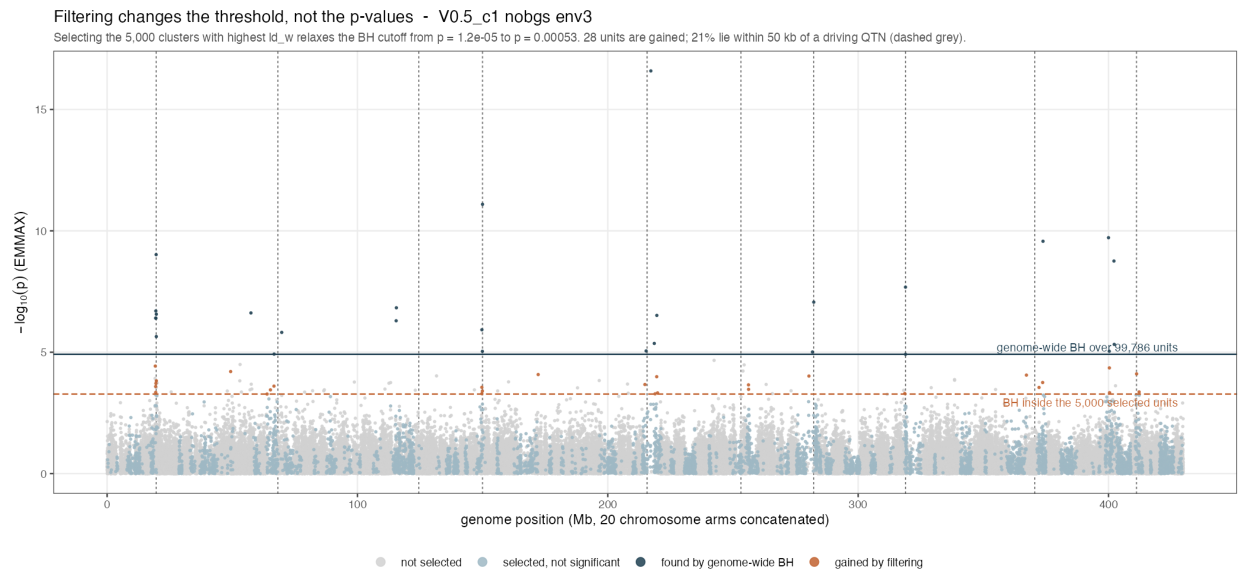

Figure 2: No p-value moves; the threshold does. Navy points clear the genome-wide cutoff, orange points lie between the two cutoffs and are gained by filtering alone.

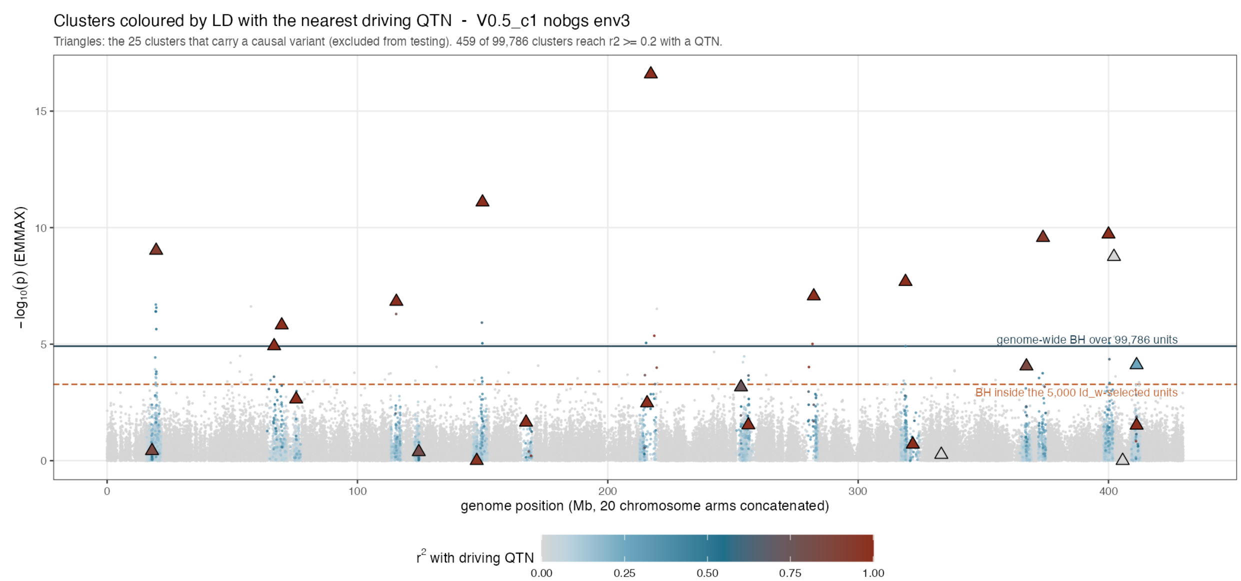

Figure 3: Clusters coloured by r^2 with the nearest driving QTN; triangles carry a causal variant.

4.4 FDR validity

Table 2: Fraction of 100 surrogate environments producing at least one rejection, averaged over 10 environments. Genome-wide baseline 0.042; alpha = 0.05. Genetic basis.

k	MAF	ld_w	random
500	0.036	0.039	0.048
1000	0.041	0.033	0.053
5000	0.028	0.027	0.043
20000	0.033	0.031	0.049
50000	0.036	0.029	0.046

Every filtered rate sits at or below α and at or below the genome-wide baseline. The baseline is honest rather than deflated: λ_{surr} runs 0.988–0.998 despite these inputs being generated with `gc_mode = "none"`.

This result is what licenses choosing k at all. Without it, any k is a fitted parameter and the power result would be worth little — the same defect that sank the C-score, whose advantage appeared only at a fitted τ.

Two cautions. False positives are heavy-tailed under ld_w: the same rejection *rate* as MAF but roughly twelve times the mean rejection *count*, so about 96% of surrogates give nothing and the remainder give tens to hundreds. Expected when selecting correlated high-LD regions, but it means an occasional wholly spurious region *set* rather than stray markers.

The `env_orth` surrogate basis is **not** usable here: it rejects genome-wide in 16–57% of surrogates while $\lambda \approx 0.99$, so it is a different-environment null rather than a no-signal null.

5 The result is not uniform across simulation cells

The pooled numbers above average over four cells that differ in selection strength and in background LD. They do not behave alike, and one of them reverses the conclusion.

Table 3: Signal available to the conventional alpha analysis, per cell (50 kb window). Where the baseline is zero, beating it requires only one discovery.

cell	panels	median true	mean true	% panels with zero
V0.5_c1	20	5.0	6.75	5
V0.5_c2	20	4.5	4.45	30
V1_c1.5	20	0.0	0.75	60
V2_c1	20	1.0	1.25	35

Table 4: Percent of 20 panels per cell in which the filter beats the genome-wide alpha analysis (50 kb window). Below 50 means the filter is WORSE than not filtering.

cell	filter	k=500	k=1000	k=2000	k=5000	k=10000	k=20000
V0.5_c1	ld_w	72	79	95	100	100	100
V0.5_c1	size	42	50	65	72	68	65
V0.5_c2	ld_w	28	28	28	28	35	100
V0.5_c2	size	100	100	100	100	100	100
V1_c1.5	ld_w	78	75	82	92	91	100
V1_c1.5	size	87	93	92	82	89	100
V2_c1	ld_w	67	75	93	100	100	100

cell	filter	k=500	k=1000	k=2000	k=5000	k=10000	k=20000
V2_c1	size	69	69	69	69	70	67

ld_w **fails in `V0.5_c2`, and cluster size does not.** ld_w wins in 28% of panels there — worse than no filtering — flat across k until the selection is so large it has stopped being a filter. Cluster size wins in **100% of panels at every** k in the same cell. This is the widest separation between the two filters anywhere in the data.

`V0.5_c2` is the high-background-LD cell ($b = 0.42$; it is the cell where the old fixed $r^2 \geq 0.27$ cut sat *below* background LD). A plausible reading is that ld_w is a **local**-LD statistic, and where everything is correlated with everything there is no local structure left for it to key on, whereas large clusters still mark real linkage blocks. That is an interpretation of one cell, not a tested mechanism.

Two consequences. First, the pooled claim in Section~6 understates the case against ld_w: the two filters are not merely indistinguishable on average, ld_w is **less robust across regimes**. Second, any application should establish which regime it is in before choosing a filter.

Table 5: Median gain in true discoveries over the genome-wide analysis, ld_w filter, 50 kb. Read alongside the win rates: a high win rate on a zero baseline can rest on a gain of one.

cell	k=500	k=1000	k=2000	k=5000	k=10000	k=20000
V0.5_c1	2.0	2.5	4.0	4.0	3.0	2
V0.5_c2	-3.5	-4.0	-3.5	-3.0	-1.5	2
V1_c1.5	0.0	0.0	0.0	1.0	0.5	0
V2_c1	0.0	1.0	1.5	1.5	1.0	1

6 The size control, and what it costs the claim

Table 6: ld_w against cluster size at identical k: percent of the 80 panels in which ld_w yields more true discoveries, with the sign-test p-value. Values near 50 percent mean the two filters are indistinguishable.

k selected	10 kb	50 kb	100 kb
500	41% (p 0.229)	44% (p 0.443)	49% (p 1.000)
1000	47% (p 0.694)	50% (p 1.000)	55% (p 0.532)
2000	55% (p 0.519)	56% (p 0.374)	58% (p 0.260)
5000	59% (p 0.220)	63% (p 0.056)	65% (p 0.030)
10000	53% (p 0.871)	63% (p 0.070)	64% (p 0.058)
20000	67% (p 0.238)	82% (p 3.2e-04)	87% (p 4.3e-06)

Selecting the k **largest** clusters performs as well as selecting on ld_w. Up to $k = 10{,}000$ the two are statistically indistinguishable at every window — win rates of 41–65% with sign-test p from 0.03 to 1.0, straddling the 50% line. This was a pre-specified control and it lands against the ld_w story across the range where filtering is most useful.

ld_w does separate at the largest selection, $k = 20{,}000$ (82% at 50 kb, 87% at 100 kb, $p \leq 4 \times 10^{-6}$). That is a real difference and is reported as such, but it is one corner of the grid, it is the regime where the filter is least selective, and the C-score failed in exactly this way — an advantage visible only at a fitted corner. It should not be promoted to the headline without a pre-specified reason to operate there.

One qualification keeps it from being fatal, and it cuts both ways. Cluster size is **not an independent rival**: the clusters exist because ld_w-based clustering built them, so size is downstream of ld_w. What the result shows is that once the clusters exist, consulting ld_w again to choose among them adds little. The information has already been spent.

7 What is established, and what is not

Claim	**Status**	**Basis**
Independent filtering of LD clusters beats a genome-wide scan	Established	80 panels, 56–97% win rate, rising with k
The filter must be LD-structure-based	Established	MAF and random are significantly WORSE than no filtering
Filtered BH controls FDR under a structure-preserving null	Established	Rates 0.027–0.039 vs α 0.05, at or below genome-wide
The gain grows with the multiplicity burden	Supported	Marker tracks (30k) weaker than pooled panels (86k); one comparison, not a designed sweep
ld_w specifically is needed at selection time	**Not supported**	Cluster size matches it at every k, and beats it outright in `V0.5_c2`
The filters behave alike across regimes	**Contradicted**	ld_w wins 28% of panels in `V0.5_c2` where cluster size wins 100%
This transfers to empirical data	Untested here	Simulations only; the 3sp panel session reports a much larger effect

7.1 What would falsify the main claim

A filter of the same size, not derived from LD structure, that matches ld_w and cluster size. MAF and recombination rate were tried and both fail badly. A positive result there would reduce the finding to arithmetic.

7.2 Known limits

The truth is bp proximity, so a discovery 49 kb from a QTN counts and one at 51 kb does not; results are reported at three windows because of this. Signal varies enormously between panels — a median of 20 significant markers per bundle, IQR 1–90, with **23% having none at all** — so single-panel figures illustrate mechanism and are not evidence. Cluster-level p-values come from one representative marker per cluster, not a within-cluster combination. Only the EMMAX engine is analysed here.

8 Provenance

8.1 Which simulation generation each result uses

These results do **not** all come from the same simulations, and the main cluster-level result does not come from the newest.

Generation	Maps	Fitted half-decay	Used for
`bgs5`	MAP_RES 3.3e-5; 31.5% of markers share a position	55.8 kb (18 win/chr, ~3 Mb spans)	**cluster-level power (the main result) and both manhattans**
`pilot_v2_v05c1`	MAP_RES 3.3e-6; no shared positions	13.3 kb (95 win/chr, ~0.56 Mb spans)	marker-level power only; `V0.5_c1` only
`analysis_inputs`	pre-bgs5	not compared	FDR validity only; `V2_c1`, nobgs, `gc_mode = none`

The two generations differ 4.2× in fitted half-decay, and at $\rho = 0.5$ the half-decay **is** the stage-2 clustering distance — so the test units themselves would differ on the newer maps. The difference traces to the decay fitting scale rather than to the maps: same-position marker pairs show r^2 of 0.023 against 0.022 for pairs 1 bp–10 kb apart, so the collapsed positions do not inflate LD. Whether the main result survives re-clustering at the newer decay settings is **untested**.

Item	Detail
Power, cluster level	`filter_then_test_clusters.R`, 80 panels x ~86k units, run on mini1 (8 cores)
Power, marker level	`filter_then_test.R`, 63 runs of `pilot_v2_v05c1` tracks
FDR validity	`filter_fdr_check.R`, 10 environments x 100 surrogates x 2 bases
Figures	`plot_filter_then_test.R`, `plot_filter_manhattan.R`, `plot_filter_manhattan_ld.R`
Bundles	`regen_sim_data_bgs5` (older generation, see above), 800 files
Surrogate p-values	`analysis_inputs`, V2_c1, B = 100, `gc_mode = none`
Tracks	`pilot_v2_v05c1/tracks`, V0.5_c1, 8 environments
Parameters	alpha 0.05; stage-2 `min_r2_rho` 0.5, `distance_threshold` 1e5; scoring `dmax_cap` 1e5
Commits	`60c8134` cap harmonisation, `88797ff` this machinery

All comparisons in this document are **paired within panel with a sign test**. Differences of means and medians are not used anywhere: three claims in this project were withdrawn after a positive mean turned out to sit on a sign count at chance.